Task 2 — Stock Portfolio Performance Tracker

Task overview and instructions

Features

• Accepts ticker symbols, buy price, current price, and share quantity

• Calculates invested capital, current market value, and per-holding gain/loss

• Calculates gain/loss percentage

• Automatically writes a formatted `portfolio_summary.csv`

Run

python portfolio.py

The script uses prices entered by the user instead of claiming to have live prices. This is a runnable educational tracker and avoids requiring a paid API.

Python source code: portfolio.py

```
"""Stock portfolio tracker with gain/loss calculations and CSV persistence."""
import csv
from decimal import Decimal, InvalidOperation
from pathlib import Path

OUTPUT_FILE = Path("portfolio_summary.csv")


def read_positive_decimal(prompt: str, allow_zero: bool = False) -> Decimal:
    while True:
        raw = input(prompt).strip()
        try:
            value = Decimal(raw)
            if value < 0 or (value == 0 and not allow_zero):
                print("Enter a positive number.")
                continue
            return value
        except InvalidOperation:
            print("Invalid number. Example: 180.50")


def collect_holding() -> dict:
    symbol = input("Ticker symbol (e.g. AAPL): ").strip().upper()
    while not symbol or not symbol.replace(".", "").replace("-", "").isalnum():
        print("Enter a valid ticker symbol.")
        symbol = input("Ticker symbol: ").strip().upper()

    buy_price = read_positive_decimal("Buy price per share: ")
    quantity = read_positive_decimal("Number of shares (decimals allowed): ")
    current_price = read_positive_decimal("Current price per share: ", allow_zero=True)
    invested = buy_price * quantity
    current_value = current_price * quantity
    gain_loss = current_value - invested
    gain_pct = (gain_loss / invested * Decimal("100")) if invested else Decimal("0")
    return {
        "symbol": symbol,
        "buy_price": buy_price,
        "quantity": quantity,
        "current_price": current_price,
        "invested": invested,
        "current_value": current_value,
        "gain_loss": gain_loss,
        "gain_pct": gain_pct,
    }


def save_csv(holdings: list[dict], filename: Path = OUTPUT_FILE) -> None:
    with filename.open("w", newline="", encoding="utf-8") as handle:
        writer = csv.writer(handle)
        writer.writerow([
```

```
            "Ticker", "Buy Price", "Quantity", "Current Price",
            "Invested Capital", "Current Value", "Gain/Loss", "Gain/Loss (%)",
        ])
        for h in holdings:
            writer.writerow([
                h["symbol"], f'{h["buy_price"]:.2f}', str(h["quantity"]),
                f'{h["current_price"]:.2f}', f'{h["invested"]:.2f}',
                f'{h["current_value"]:.2f}', f'{h["gain_loss"]:.2f}',
                f'{h["gain_pct"]:.2f}',
            ])
        total_invested = sum((h["invested"] for h in holdings), Decimal("0"))
        total_value = sum((h["current_value"] for h in holdings), Decimal("0"))
        total_gain = total_value - total_invested
        total_pct = total_gain / total_invested * Decimal("100") if total_invested else
    Decimal("0")
        writer.writerow([])
        writer.writerow(["TOTAL", "", "", "", f"{total_invested:.2f}",
                         f"{total_value:.2f}", f"{total_gain:.2f}", f"{total_pct:.2f}"])


def main() -> None:
    print("\n=== STOCK PORTFOLIO PERFORMANCE TRACKER ===")
    print("Enter the prices you want to track. These calculations are not live market data.")
    holdings: list[dict] = []
    while True:
        holdings.append(collect_holding())
        if input("Add another holding? (y/n): ").strip().lower() not in {"y", "yes"}:
            break

    print("\nPORTFOLIO SUMMARY")
    print("-" * 78)
    print(f'{"Ticker":10} {"Invested":>12} {"Current":>12} {"Gain/Loss":>12} {"Gain %":>10}')
    print("-" * 78)
    for h in holdings:
        print(f'{h["symbol"]:10} {h["invested"]:>12.2f} {h["current_value"]:>12.2f} '
              f'{h["gain_loss"]:>12.2f} {h["gain_pct"]:>9.2f}%')
    total_invested = sum((h["invested"] for h in holdings), Decimal("0"))
    total_value = sum((h["current_value"] for h in holdings), Decimal("0"))
    total_gain = total_value - total_invested
    total_pct = total_gain / total_invested * Decimal("100") if total_invested else
    Decimal("0")
    print("-" * 78)
    print(f"TOTAL INVESTED: {total_invested:.2f}")
    print(f"CURRENT VALUE:  {total_value:.2f}")
    print(f"TOTAL GAIN/LOSS:{total_gain:.2f} ({total_pct:.2f}%)")
    try:
        save_csv(holdings)
        print(f"\nSaved CSV report to: {OUTPUT_FILE.resolve()}")
    except OSError as exc:
        print(f"Could not save CSV report: {exc}")


if __name__ == "__main__":
    try:
        main()
    except (KeyboardInterrupt, EOFError):
        print("\nTracker stopped.")
```